6.12 Improper Integrals

Name _____ Period ____ Date _____

AP Calculus 6.12 · Student sheet

I can…

- write an improper integral as a limit.
- decide whether an improper integral converges or diverges.
- evaluate improper integrals with infinite limits or an infinite discontinuity.

Vocabulary · fill in the missing word

Improper integral — an integral with an _____ limit of integration or an integrand that is not defined at a point of [a, b]

Converges — the limit _____ and is a real number

Diverges — the limit does not exist or is _____

Part 1 · Infinite region, finite area?

Areas from $x = 1$ to $x = b$.

b	10	100	1000
$\int_1^b dx/x^2 = 1 - 1/b$			
$\int_1^b dx/x = \ln b$			

Part 2 · Notes and examples (fill in each note, then work the example)

Infinite limit: ∫ from a to ∞ of f(x) dx = lim (b → ∞) ∫ from a to b; it converges if the _____ exists.

Example 1

Evaluate ∫ from 1 to ∞ of dx/x^2.

p-integrals: ∫ from 1 to ∞ of dx/x^p converges when $p >$ _____ and diverges when $p \le 1$.

Example 2

Determine whether ∫ from 1 to ∞ of dx/x converges.

Infinite discontinuity: if f is not defined at an endpoint, use a one-sided _____.

Example 3

Evaluate ∫ from 0 to 1 of $dx/\sqrt{x}$.

Example 4

Evaluate $\int$ from 0 to ∞ of $e^{-2x}\,dx$.

Example 5

Evaluate $\int$ from $-\infty$ to ∞ of $dx/(1 + x^2)$.

Part 3 · Partner practice

Partner 1

Evaluate $\int$ from 1 to ∞ of dx/x^3.

Partner 2

(AP® Practice) $\int$ from 1 to ∞ of $\frac{2}{x^3}\,dx$ =

(A) 1 (B) $\frac{1}{2}$ (C) 2 (D) diverges